

INFRASTRUCTURE RESOURCE REPORT — PHASE 2

Infrastructure Report — Phase 2

Events + Social Seed

v2.0 · Richard Uzor — May 2026 · Devstrike Digital Limited

01 — PHASE 2 INFRASTRUCTURE CHANGES

Phase 2 upgrades the infrastructure to match growing traffic and introduces new components for payments, background jobs, push notifications, and the PWA. Monthly cost rises to ~~~₦~~\$130,000 but is more than covered by Phase 2 revenue.

02 — UPGRADES FROM PHASE 1

Frontend Hosting *Vercel Pro*

Tier: Pro (\$20/month)

Estimated Cost: ~~₦~~\$32,000/month

Upgrade from Hobby. Needed for: team deploys without queue, 1TB bandwidth, preview deployments per PR, advanced analytics, image optimisation.

Scaling trigger: Already at this tier if Phase 1 traffic exceeded free bandwidth. No further upgrade in Phase 2.

Backend API + Database *Railway Team (paid)*

Tier: ~\$15–25/month

Estimated Cost: ~~¥24,000–¥40,000/month~~

Dedicated PostgreSQL instance replaces shared. API container gets dedicated resources. PgBouncer connection pooling enabled. Automated daily backups with 30-day retention.

Scaling trigger: Upgrade database to 10GB storage tier if approaching 8GB used.

03 — NEW PHASE 2 COMPONENTS

Background Job Worker *Railway (second container)*

Tier: ~\$5–10/month

Estimated Cost: ~~¥8,000–¥16,000/month~~

Runs BullMQ worker process. Handles: Paystack webhook processing, email notifications (decoupled from API response), quote card server-side generation, report threshold checks.

Scaling trigger: Scale to multiple worker instances if job queue depth consistently > 100 pending jobs.

Cache Layer *Upstash Redis*

Tier: Free tier (10k commands/day)

Estimated Cost: ~~¥0/month~~ (Phase 2)

Rate limiting counters, hot feed data (5-min TTL), reaction counts, session tokens. Prevents DB hit on every page load for popular content.

Scaling trigger: Upgrade to paid when daily commands > 8,000.

Push Notifications *Firebase Cloud Messaging*

Tier: Free

Estimated Cost: ~~¥0/month~~

Sends push to PWA-installed users. Event reminders, 'pastor posted', 'question answered'. FCM has no practical limit for Phase 2 scale.

Scaling trigger: No upgrade required until millions of daily notifications.

Calendar Integration *Google Calendar API*

Tier: Free (quota-based)

Estimated Cost: ₦0/month

Google Calendar API is free within the quota (1 million requests/day). Phase 2 calendar saves will be a tiny fraction of this.

Scaling trigger: No upgrade trigger in Phase 2 or 3.

04 — PHASE 2 COST SUMMARY

Item	Monthly Cost
Vercel Pro	₦32,000
Railway (API + DB + Worker)	₦40,000–₦60,000
Cloudflare R2	₦8,000–₦16,000
Upstash Redis	₦0
Resend (Pro)	₦32,000
FCM	₦0
Plausible	₦14,400
Sentry	₦0
Misc + buffer	₦10,000
TOTAL	₦136,400–₦164,400/month